# UI-Mem: Self-Evolving Experience Memory for Online Reinforcement Learning in Mobile GUI Agents

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Paper version: arXiv 2602.05832v1 · CUA Radar Reel poster, not made or endorsed by the authors

## Problem

> *Online RL for mobile GUI agents has sparse rewards and repeats the same errors across tasks.*

Replay buffers and dense rewards do not carry **reusable experience** across tasks or apps.

## Motivation: three gaps in online RL

1. **Credit assignment.** A final error erases partial progress.
2. **Repeated errors.** Same failures rediscovered.
3. **No transfer.** Replay and dense rewards stay in-task.

## Evaluation setup

AndroidWorld · 116 tasks | AndroidLab · 138 tasks | Qwen3-VL 4B / 8B | 256 training queries | G = 4 rollouts | 5 epochs, lr 1e-6

- **All online** emulator benchmarks; no offline grounding accuracy.
- **Held out:** 5 AndroidLab apps; model size, tasks per app, seeds not stated.

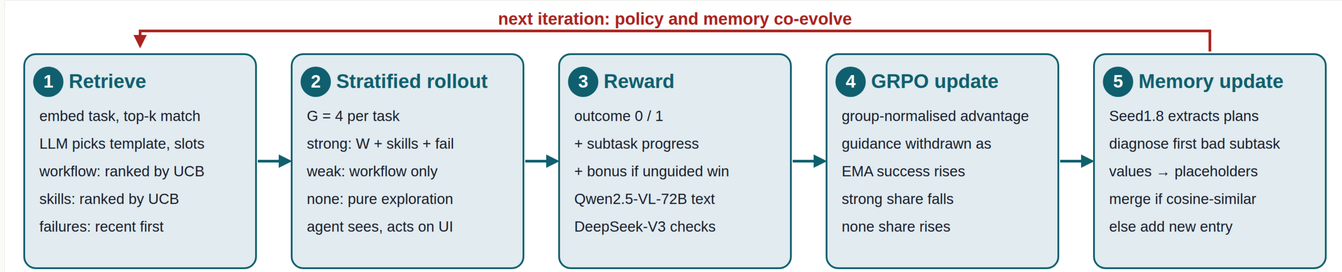
## Our read (CUA Radar)

| 76 | ★★★ | transfer |
|---|---|---|
| score of 100 | band | evidence level |

- **Reports:** online eval, held-out apps, ablation, judge check. **Not reported:** seeds, matched budget.
- **Flags:** possible memory/data overlap with eval apps (AP5).
- Pattern C19. Authors' numbers, not rerun.

## Method: hierarchical memory steers stratified GRPO rollouts

- **Sees / acts:** screenshot + instruction in; Click, Swipe, Type, Home, open_app out.
- **Memory:** workflows, subtask skills and failure patterns as slot templates, UCB-ranked.
- **Signal:** outcome + subtask progress + bonus for unguided success, scored by a text-based LLM judge.

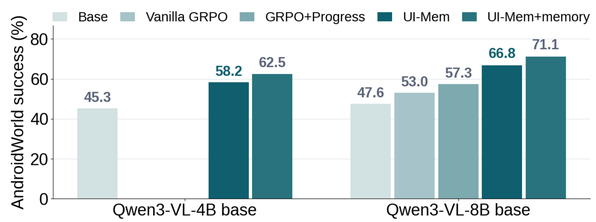



*Our redraw (not the paper's figure): the retrieve, rollout, reward, update and memory loop, with the three memory levels.*

$$R(\tau) = \lambda_o\, r_{\text{outcome}} + \lambda_p\, r_{\text{progress}} + \alpha\, \mathbb{I}[\text{guidance}(\tau) = \text{None}]\, r_{\text{outcome}}$$

*guidance-aware reward (paper Eq. 5)*

## Key results: AndroidWorld online success (%)




*Tables 1, 3 redrawn. Online, 116 tasks; "+memory" = test-time retrieval. Not drawn: GRPO + Experience Replay 56.5.*

**8B: 66.8 / 71.1 vs 47.6 base and 53.0 Vanilla GRPO**

**SO WHAT →** *Same Qwen3-VL bases; the paper gives GRPO baselines for 8B only. AndroidLab SR: 37.7 (4B), 43.5 (8B). Training queries: 256, seeded from AMEX, AndroidLab, UI-Genie and augmented with GPT-4o.*

## Ablation: AndroidWorld, 8B (%)

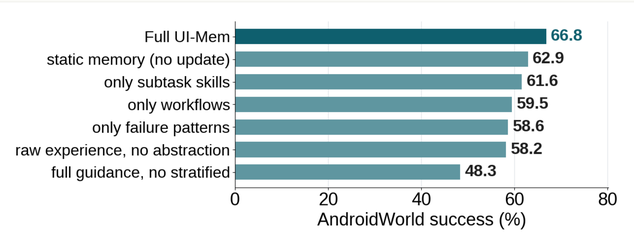



*Figure 5, redrawn: one component removed or replaced at a time, all on AndroidWorld with the 8B model.*

**SO WHAT →** *Removing stratified sampling (full guidance) gives 48.3, below Vanilla GRPO at 53.0 and test-time prompting alone at 52.6. Raw experience instead of templates gives 58.2; one memory level only gives 58.6 to 61.6 (59.5 for workflows). No variance reported. Disabling the memory update gives 62.9 against 66.8 for the full system, so the self-evolving loop is part of the gain. The paper states no seeds and no confidence intervals for any variant.*

## Headline Numbers

**71.1%**
**AndroidWorld, 8B**
*with test memory; base 47.6%*

- **66.8%** 8B, no test memory
- **58.2%** 4B, no test memory
- **43.5%** AndroidLab SR, 8B

## The paper's own controls

| CONTROL (ONLINE) | UI-MEM | COMPARED WITH |
|---|---|---|
| AndroidWorld, 8B | 66.8 | Vanilla GRPO 53.0 |
| Held-out Maps.me, 0-shot/mem | 33.3 / 33.3 | base 13.3 |
| Held-out Zoom, 0-shot/mem | 40.0 / 80.0 | base 40.0 |
| Judge accuracy | 0.900 | MLLM-only 0.724 |

## Takeaway

**Memory-guided GRPO: 8B 47.6% to 66.8% on AndroidWorld (71.1% with test memory); no variance.**

AndroidLab 8B SR 43.5% | MobileRL 42.5% | judge F1 0.902

## Scan to Read

Paper | Project